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BLDC Fan Controller with Current Control



The BLDC ceiling fan controller reference design features sensorless sinusoidal current control, converting 90-265V AC to 24V DC. Designed for industrial applications, it offers precise speed regulation and efficiency.

Texas Instruments (TI) introduces a brushless direct current (BLDC) ceiling fan controller with sensorless sinusoidal current control, the TIDA00386 reference design. Operating from AC power, it converts 90-265V AC to 24V DC, offering a comprehensive solution for ceiling fan applications with a 3-phase BLDC motor, suitable for industrial applications like dental chairs and power conditioning systems. The design features sinusoidal sensorless control at 25kHz for minimal torque ripple and reduced audible noise, operating within a wide universal AC voltage range. The tightly regulated 24V DC power supply ensures $\pm 1\%$ speed regulation across the input voltage range, with only 0.6% ripple on the 24V DC output for optimised acoustic performance. The AC/DC section achieves high efficiency exceeding 85% under all load and line conditions, meeting stringent energy efficiency standards. Comprising three sections - AC-DC, motor control, and IR remote-based speed control - the design employs a flyback AC/DC topology for the universal main AC input range, an IC for 3-phase BLDC motor control, and a MSP430G2201 microcontroller for IR remote-based fan speed control.

The reference design is suitable for ceiling fan applications utilising a 3-phase BLDC motor.

Texas Instruments (TI)

<https://www.electronicshobby.com/electronics-projects/electronics-design-guides/reference-design-for-blcd-fan-controller-with-current-control>

Low Voltage Ceiling Fan



A low-voltage ceiling fan reference design offers energy-efficient operation at 300RPM, consuming 17.5W, equivalent to two 60W LED bulbs. It controls a 3-phase motor and integrates advanced features for effective motor control and energy savings.

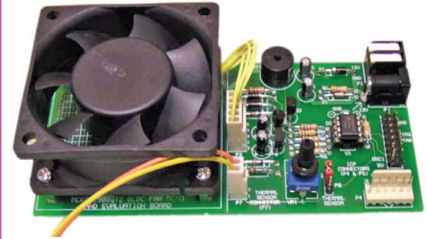
Microchip's reference design for a low-voltage ceiling fan offers a cost-effective and energy-efficient solution for a low-voltage PMSM fan, operating at 300RPM while consuming approximately 17.5W, equivalent to the energy required by two 60W LED bulbs. This design controls a low-voltage, three-phase PMSM/BLDC ceiling fan motor within a 90-265V AC input range, driving motors up to 40W. The design integrates features like a three-phase motor control power stage, a 32V, 45W flyback power supply with switched valley-fill PFC, and an auxiliary power supply for on-board circuitry. It includes DC bus current and voltage feedback for protection and control, phase voltage feedback for sensorless BEMF control, and uses a dsPIC33CK64MC102 DSC with enhanced peripherals for high-speed PWM, ADC, and communication interfaces. The power conversion circuit employs a flyback converter for input power transformation and a PWM DC/DC converter for voltage regulation, ensuring energy efficiency with a power factor above 0.9.

The reference design is suitable for ceiling fan applications utilising a 3-phase BLDC motor.

Microchip

<https://www.electronicshobby.com/electronics-projects/electronics-design-guides/reference-design-for-low-voltage-ceiling-fan>

BLDC Fan Controller



The BLDC fan controller reference design provides efficient thermal speed control, focusing on noise reduction, energy efficiency, and fault protection. It includes an 8-pin MCU with auto-thermal speed adjustment, motor-lock protection, and firmware upgrades.

NXP's reference design BLDC fan controller provides an efficient solution for implementing thermal speed control in BLDC fans, focusing on noise reduction, energy efficiency, thermal management, and fault protection. Powered by the 8-pin MC68HC908QT2 microcontroller unit (MCU), the design offers features such as auto-thermal speed adjustment, motor-lock protection, alarms for over-temperature and motor-lock conditions, in-circuit programming for firmware upgrades, and manual speed adjustment. The design comprises a BLDC fan, Hall effect sensor, MCU, regulator, transistors, thermal sensor, and buzzer, with the MCU driving the fan by controlling the transistors based on signals from the Hall effect sensor for commutation. The fan's speed is determined by an external thermal sensor, with the MCU adjusting the off-time period accordingly. This design's features, including a regulated 12V, 1A power input, a piezoelectric buzzer for motor-lock alarm, and motor-lock protection circuitry, ensure reliable operation and customisable thermal speed profiles for diverse cooling requirements.

The reference design presents a solution for implementing thermal speed control in BLDC fans.

NXP

<https://www.electronicshobby.com/electronics-projects/industry-grade-projects/reference-design-for-blcd-fan-controller>